

VYSHNAVI PRIYA KASARLA

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SUMMARY

Data Analyst & Data Scientist with 1+ year of experience in healthcare data, SQL-based data quality management, and machine learning. Skilled in SQL (BigQuery, PostgreSQL), Python (pandas, scikit-learn, TensorFlow), Power BI, Tableau, ETL pipelines, and FHIR/HL7 interoperability. Built ML models reaching 96.4% accuracy and analytics dashboards that turn complex data into clear recommendations for technical and non-technical stakeholders. Google Data Analytics Professional Certificate holder.

EDUCATION

- MS in DataScience | Michigan Technological University | GPA 3.8/4.0 Aug 2024 - Apr 2026
- B-Tech in Computer Science | Rajiv Gandhi University of Knowledge and Technology | GPA 8.8/10 Jun 2020 - May 2024

EXPERIENCE

Graduate Research Assistant - Public Health Disease Surveillance (Funded) May 2025 - Present

Biomedical Data Science Lab, Michigan Technological University Houghton, MI

- Developed FHIR interoperability models creating longitudinal patient records from 3,000+ HL7 sources performed data mapping, data analysis, and SQL-based quality management to detect disease hotspots and generate analytical reports for health plan management.
- Builds and maintains secure Next.js/Node.js information systems with real-time dashboards, CI/CD pipelines, and AI/ML models for disease prediction, cutting manual reporting time by 60%; gathered requirements from UPHCS clinical departments, delivered user training, and produced technical documentation.

Graduate Research Assistant - K-12 Mental Health Analytics Platform Oct 2025 - Present

Biomedical Data Science Lab, Michigan Technological University Houghton, MI

- Develops and maintains a mental health analytics platform integrating PHQ-9, GAD-7, and SDOH screening tools with ML risk scoring, REST APIs, and React dashboards for 4 user roles; documents requirements, service delivery standards, and duties throughout.
- Integrating blockchain-based data architecture to ensure tamper-proof storage and secure exchange of sensitive student mental health records across school districts, enhancing data integrity, privacy compliance, and auditability throughout the platform's information systems.

Teaching Assistant - Population Health Informatics Jan 2026 - Apr 2026

Michigan Technological University, Houghton, MI

- Supported instruction in healthcare information systems, data standards, EHRs, biosurveillance, data quality management, epidemiological GIS, and health plan policies across multiple services and departments.
- Maintained grading records, provided technical feedback, and developed training documentation supporting student knowledge of regulatory standards and clinical information management.

PROJECTS

Medicare Claims Cost Analysis - Google Data Analytics Capstone ,SQL (BigQuery), Power BI Aug2026 - Sep 2026

- Cleaned, joined, and reconciled 3 relational tables (138,556 members, 558,211 claims) in BigQuery SQL, using window functions to flag 30-day readmissions and validating all totals to the dollar (\$556.5M). Found that 5% of members drove 44% of cost and members with 5+ chronic conditions drove ~65% of spending; built a Power BI dashboard translating findings into care-management recommendations.

Energy-Efficient Deep Learning - Predictive GPU/CPU Modeling. Jan 2026 - Mar 2026

- Addressed a critical gap in sustainable AI by predicting GPU/CPU energy consumption of deep learning jobs before execution, enabling efficient scheduling and reducing carbon emissions. Benchmarked 9 regression models on 29,647 NREL BUTTER-E workload records; Neural Network achieved the best performance ($R^2=0.9974$, $RMSE=0.0498$), with nonlinear models outperforming all linear baselines.

Multimodal Fetal Abnormality Detection - AI/ML | Team Lead (Group of 2). Dec 2024 - Apr 2025

- Developed an AI system for early detection of fetal abnormalities using a multimodal approach combining ultrasound images and non-invasive ECG signals to identify brain malformations (CSP, LV) and cardiac arrhythmias through real-time automated predictions; achieved 96.4% accuracy and 0.97 F1-score with SHAP/Grad-CAM applied for clinical interpretability.

TECHNICAL SKILLS

Data & SQL : SQL, BigQuery, PostgreSQL, MySQL, ETL, Data Mapping, Data Quality Management, FHIR/HL7, Apache Spark
Programming : Python (pandas), R, JavaScript, HTML5, CSS
BI & Reporting : Power BI, Tableau, Excel, Google Sheets, Recharts
ML & AI : scikit-learn, TensorFlow, Keras, Random Forest, SVM, SHAP, AI Agent Frameworks (MCP)
Web & Engineering: React, Next.js, Node.js, FastAPI, REST APIs, Docker, CI/CD, Git

CERTIFICATIONS

- Google Data Analytics Professional Certificate - Google (Sep 2026)
- Microsoft Power BI - Microsoft Learn (Mar 2026)
- The Complete SQL Bootcamp - Udemy (2026) | Data Science - Udemy (2024)